DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Mawlana Bhashani Science and Technology University, Santosh, Tangail-1902

CLASS ROUTINE : EFFECTIVE FROM 26 SEPTEMBER, 2026

THIRD YEAR FIRST SEMESTER

Time	09:00-09:50	10:00-10:50	11:00-11:50	12:00-12:50	12:50-01:40	01:45-02:25	02:30-03:10	03:15-03:40
Saturday					BREAK	ICT 3101 NI	ICT 3107 ZR	
Sunday		ICT 3108 Lab ZR		ICT 3103 MAK			ICT 3109 MH	
Monday		ICT 3107 ZR		ICT 3105 MAH		ICT 3101 NI	ICT 3102 Lab NI	
Tuesday	ICT 3103 MAK		ICT 3105 MAH			ICT 3101 NI	ICT 3104 Lab MAK	
Wednesday			ICT 3106 Lab MAH			ICT 3109 MH		

Course Code	Course Name	Faculty
ICT 3101	Data Communication –	NI
ICT 3102	Data Communication Lab	– NI
ICT 3103	Microwave Engineering	MAK
ICT 3104	Microwave Engineering Lab	MAK
ICT 3105	Database Management System	MAH
ICT 3106	Database Management System Lab	MAH
ICT 3107	Software Engineering & Project Management	ZR
ICT 3108	Software Engineering & Project Management Lab –	ZR
ICT 3109	Theory of Computation –	MH

WEEKLY STUDY PLAN (SUBJECT FOCUS)

Day	Main Focus
Saturday	Data Communication + Microwave Engineering
Sunday	Database Management System
Monday	Software Engineering & Project Management
Tuesday	Microwave Engineering + Data Communication
Wednesday	Theory of Computation
Thursday	Weekly Revision + Assignment + Weak Topic
Friday	Project + Coding + Backlog Clear

THURSDAY – ACADEMIC DEEP WORK

08:00–10:00	Difficult Subject
10:30–12:00	Database / Theory
02:00–04:00	Lab / Assignment
05:00–06:30	Coding
08:00–09:30	Weekly Revision

FRIDAY – PROJECT & GROWTH DAY

08:00–10:00	Project Work
10:30–12:00	Coding
02:30–04:00	Weak Topic Clear
05:00–06:00	Exercise / Sports
07:00–08:00	English / Communication Skill
08:30–10:00	Next Week Preparation

SATURDAY

07:00	Wake up
08:00–11:30	Study / Skill
01:45–03:10	Class
04:30–06:00	Revision
06:30–07:30	Coding
08:00–09:30	Assignment
11:00	Sleep

SUNDAY

10:00–11:50	Class/Lab
12:00–12:50	Class
01:45–03:10	Class
04:30–05:30	Revision
06:00–07:00	Coding
08:00–09:30	Lab / Assignment
11:00	Sleep

MONDAY

10:00–11:50	Class
12:00–12:50	Class
01:45–03:10	Class
04:30–05:30	Database Study
06:00–07:00	Coding
08:00–09:30	Assignment
11:00	Sleep

TUESDAY

09:00–09:50	Class
11:00–12:50	Class
01:45–03:10	Class
04:30–06:00	Deep Study
06:30–07:30	Coding
08:00–09:30	Lab / Assignment
11:00	Sleep

WEDNESDAY

08:00–09:30	Deep Study
11:00–12:50	Class/Lab
01:45–03:40	Class
05:00–06:30	Project Work
07:00–08:00	Coding
09:00–10:00	Weekly Revision
11:00	Sleep

Note:
Tuesday wake up **6:30 AM**
(First class at 09:00 AM)

DAILY EFFECTIVE ROUTINE (MONDAY TO FRIDAY)

07:00 AM	Wake up
07:00–07:20	Fresh + Water
07:20–07:40	Exercise / Walking
07:40–08:10	Breakfast
08:10–08:40	Quick Revision (Previous Day)
08:40–09:00	Preparation / Go to University
09:00 AM – Class Time	University Class (As per Routine)
During Break	Water + Light Snack + Short Break
After Class	Lunch
03:30–04:00	Return Home + Rest
04:00–04:30	Power Nap / Relax
04:30–06:00	Main Study Session
06:00–06:30	Break + Walking
06:30–07:30	Coding / Skill Development
07:30–08:00	Dinner
08:00–09:30	University Study / Assignment
09:30–10:00	Free Time
10:00–10:30	Revision + Next Day Preparation
10:30–11:00	Phone Off + Relax
11:00 PM	Sleep (8 Hours)

7 GOLDEN RULES FOR SUCCESS

1. 11 PM-এর মধ্যে ঘুম — 8 Hours Sleep নিশ্চিত করুন।
2. প্রতিদিন minimum 3 Hours Self-Study করুন।
3. 50–10 Rule: 50 min study → 10 min break।
4. Class-এর 24 ঘন্টার মধ্যে Revision করুন।
5. প্রতিদিন 30–60 min Coding করুন।
6. Thursday/Friday Backlog Clear করুন।
7. Phone Control: Study time-এ Phone Silent + Away রাখুন।

DAILY IDEAL TARGET

- [x] University Class
- [x] 2–3 Hours Academic Study
- [x] 30–60 min Coding / Project
- [x] 20–30 min Exercise / Walking
- [x] 8 Hours Sleep

Remember!

প্রতিদিন 3–4 ঘন্টা Focused Work করাই Success-এর মূল চাবিকাঠি।

Plan Your Day Work Smart

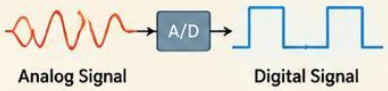
Consistency Today Success Tomorrow

Stay Focused Keep Growing

ICT 3101 Data Communication — 3 Credits

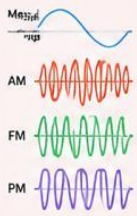
1 Introduction to Analog and Digital Communication

- Bandwidth
- Signal-to-noise ratio (SNR)
- Capacity



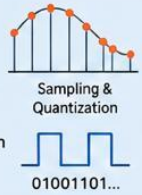
2 Analog Modulation

- Amplitude modulation (AM)
- Double sideband suppressed carrier (DSB-SC)
- Single sideband (SSB)
- Vestigial sideband (VSB)
- Frequency modulation (FM)
- Phase modulation (PM)



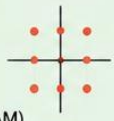
3 Analog-to-Digital Conversion

- Sampling
- Quantization
- Pulse code modulation (PCM)
- Differential PCM (DPCM)
- Adaptive DPCM (ADPCM)
- Delta modulation
- Digital telephony and video compression
- Line coding
- Pulse shaping
- Intersymbol interference (ISI)



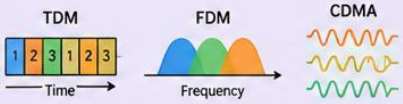
4 Digital Modulation

- Amplitude shift keying (ASK)
- Frequency shift keying (FSK)
- Phase shift keying (PSK)
- Differential, binary and M-ary PSK
- Quadrature amplitude modulation (QAM)
- Binary and M-ary FSK
- Minimum shift keying (MSK)
- Gaussian minimum shift keying (GMSK)



5 Multiplexing and Multiple Access

- Time division multiplexing (TDM)
- Frequency division multiplexing (FDM)
- Orthogonal frequency domain multiplexing (OFDM)
- Spread spectrum
- Code division multiple access (CDMA)



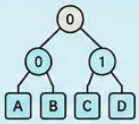
6 Noise in Communication

- Noise in communication
- Additive white Gaussian noise (AWGN)
- Errors due to noise



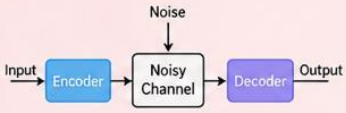
7 Information Theory

- Entropy
- Mutual information
- Huffman coding
- Lempel-Ziv source coding
- Shannon's information capacity theorem



8 Error Control Coding

- Linear block codes
- Cyclic codes
- Convolutional codes
- Viterbi decoding



9 Modern Communication Systems

- Cellular communication
- Satellite communication
- Optical fiber communication
- Power-line carrier communication



Reference Books

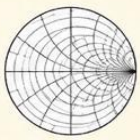
1	Data Communication: A Top-down Approach	J.F. Kurose, K.W. Ross
2	Data Networks	Bertsekas and Gallagar
3	Computer Networks	Andrew S. Tanenbaum
4	Communication Networks	Sharam Hekmat

ICT 3103 Microwave Engineering — 3 Credits

1 UHF Transmission Lines

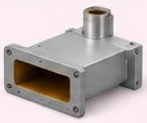
- Voltage and current in ideal transmission lines
- Reflection and transmission
- Standing wave
- Impedance transformation
- Smith chart
- Impedance matching
- Lossy transmission lines





2 Microwave Components

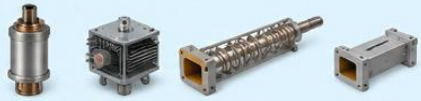
- Cavities
- Slow wave structures
- Waveguide Tees
- Directional Couplers
- Circulators and Isolators
- S-parameters





3 Microwave Tubes

- Klystron amplifier
- Multicavity klystron amplifier
- Reflex klystron oscillator
- Magnetron
- TWT (Traveling Wave Tube) amplifier
- BWO (Backward Wave Oscillator)



4 Semiconductor Microwave Devices

- Tunnel diodes
- Gunn-Effect diodes
- IMPATT diodes
- Microwave measurements



5 Antennas

- Small current element
- Radiation resistance
- Radiation pattern and properties
- Hertzian and half wave dipoles
- Monopole
- Horn antenna
- Rhombic and parabolic reflector
- Array antenna
- Yagi-Uda antenna



6 Radar and Electronic Navigation Systems

- Radar principles
- Civil, military and weather applications
- Radar equation
- Transmitters and radiators
- Factors influencing maximum range
- Moving target indicator (MTI)
- Tracking radar system and search systems
- Electronic navigation systems
- LORAN, ILS, SONAR



7 Modern Trends in Microwaves Engineering

- Effect of microwaves on human body
- Medical and civil applications of microwaves
- Electromagnetic interference / Electromagnetic Compatibility (EMI / EMC)
- Monolithic Microwave IC fabrication
- RFMEMS for microwave components
- Microwave Imaging



Reference Books

1	Foundations for Microwave Engineering	R.E. COLLIN
2	Microwave Engineering	D.M. POZAR
3	Antennas	J.D. KRAUSS
4	Field and Wave Electromagnetics	D.K. CHENG

ICT 3105 Database Management System — 3 Credits

1 Concepts of Database Systems

- What is a database?
- Database systems and DBMS
- Characteristics and advantages
- Data independence
- Users and DBA
- Database architecture
- Applications of DBMS



2 Relational Model

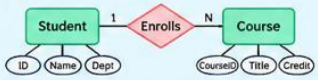
- Introduction to the relational model
- Relations, tuples, attributes
- Domain, degree, cardinality
- Keys and integrity rules

Student

ID	Name	Dept	CGPA
1	Rahim	CSE	3.75
2	Karima	EEE	3.60
3	Tanvir	CSE	3.85

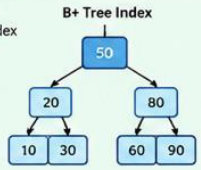
3 Database Design (ER Model)

- Entity, attribute, relationship
- E-R diagrams
- Business rules
- Mapping constraints
- Weak entity, strong entity
- Database design process



4 Indexing

- Primary index
- Secondary index
- B+ tree index
- Hash index



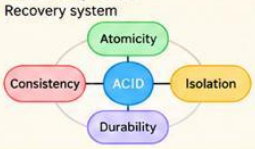
5 Query Processing and Optimization

- Query processing steps
- Query optimization techniques
- Query evaluation
- Relational algebra operations
- Cost-based optimization



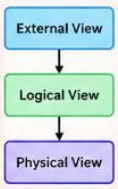
6 Transaction Management

- Transactions and ACID properties
- Transaction states
- Schedule and serializability
- Concurrency control
- Recovery system



7 Data Models

- Basic building blocks
- Business rules
- The evolution of data models
- Degrees of data abstraction
- E-R diagrams (review)



8 Relational DBMS

- Logical view of data
- Keys and integrity rules
- Selection and projection
- Set operations
- Renaming
- Joins (inner, outer, etc.)
- Division
- Syntax and semantics
- Operators, grouping and ungrouping
- Relational comparison
- Atomic domain and normalization
- Multi-valued dependency
- Join dependency

9 Relational Calculus

- Tuple relational calculus
- Domain relational calculus
- Calculus vs algebra
- Computational capabilities
- Data storage structures

Tuple Relational Calculus:

$\{ t \mid \text{Student}(t) \wedge t.\text{Dept} = \text{'CSE'} \}$

Domain Relational Calculus:

$\{ \langle x, y \rangle \mid \text{Student}(x, y, z) \wedge z = \text{'CSE'} \}$

10 Parallel and Distributed Databases

- Need for parallel and distributed databases
- Architectures and data distribution
- Transaction management in distributed DDBS
- Distributed query processing
- Challenges and advantages



11 Big Data Analytics & NoSQL

- Introduction to big data analytics
- Characteristics of big data
- NoSQL databases
- Types of NoSQL (key-value, document, column, graph)
- Applications and comparisons



12 Focus on Program: SQL

- DDL (CREATE, ALTER, DROP)
- DML (INSERT, UPDATE, DELETE)
- DQL (SELECT, WHERE, GROUP BY, HAVING)
- Joins and subqueries
- Jiews, indexes
- Stored procedures and functions
- Transaction control (COMMIT, ROLLBACK)

```
SELECT s.Name, c.TItle
FROM Student s
JOIN Enroll e ON s.ID = e.StudentID
JOIN Course c ON e.CourseID = c.CourseID
WHERE s.Dept = 'CSE'
GROUP BY s.Name, c.Title;
```

Summary

- Covers database concepts to advanced topics
- Includes design, querying, transactions, concurrency and recovery
- Focuses on relational model, normalization and calculus
- Introduces distributed databases, big data and NoSQL
- Practical SQL programming emphasis

Reference Books

1	Database Management Systems	Carlos Colonel, Steven Morris
2	Readings In Database Systems	Joseph M. Hellerstein, Michael Stonebraker
3	Fundamentals of Database Systems	Ramez Elmasri, Sham Navathe
4	Practical SQL: A Beginner's Guide to Storytelling with Data	Anthony DeBarros

ICT 3107 Software Engineering and Project Management — 3 Credits

1 Introduction

- The evolving role of software
- Software and its characteristics
- Changing nature of software
- Software engineering
- A process framework
- Process patterns
- Process assessment
- Personal and team process models
- Process technology
- Product and process



2 Process Models

- Prescriptive models
- Waterfall model
- Incremental process models
- Evolutionary process models
- Specialized process models



3 Requirements Engineering

- Requirements engineering task
- Initiating the requirements engineering process
- Eliciting requirements
- Developing use cases
- Building the analysis model
- Negotiating requirements
- Validating requirements
- Software requirement document (SRS)

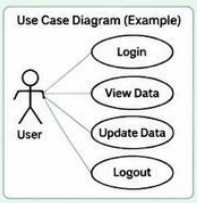
SRS (Software Requirement (Specification))



Requirements Engineering Process

4 Building the Analysis Models

- Requirement analysis
- Analysis model approaches
- Data modeling concepts
- Object oriented analysis
- Scenario-based modeling
- Flow-oriented modeling
- Class based modeling
- Creating a behavioral model



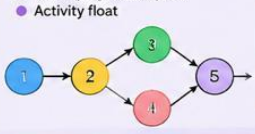
5 Introduction to Project Management

- Introduction
- Project and importance of project management
- Contract management
- Activities covered by software project management
- Plans, methods and methodologies
- Stakeholders
- Setting objectives
- Business case
- Success and failure
- Management and management control
- Project management life cycle
- Traditional vs modern project management practices



6 Activity Planning

- Objectives of activity planning
- When to plan
- Project schedules
- Sequencing and scheduling activities
- Network planning models
- Forward pass – Backward pass
- Identifying critical path
- Activity float



Critical Path Example

Critical Path: 1 → 2 → 3 → 5

7 Software Quality

- Introduction
- The place of software quality in project planning
- Importance of software quality
- Software quality models
- ISO 9126
- Quality management systems
- Process capability models
- Techniques to enhance software quality
- Quality plans



8 Software Testing

- A strategic approach to software testing
- Strategic issues
- Test strategies for conventional software
- Test strategies for object-oriented software
- Validation testing
- System testing
- The art of debugging



Reference Books

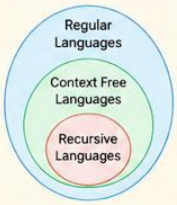
1	Object Oriented Modelling and Design with UML	Michael Blaha, James Rumbaugh
2	Software Project Management	Bob Hughes, Mike Cotterell
3	Software Engineering – A Practitioners Approach	Roger S. Pressman
4	Software Engineering	Ian Sommerville

ICT 3109 Theory of Computation

3 Credits

1 Regular Languages

- Alphabet, strings and languages
- Regular languages and operations
- Closure properties
- Regular expressions and finite automata
- Applications of regular languages



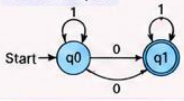
Regular Language Example

L = { w | w over {0,1} containing even number of 0's }

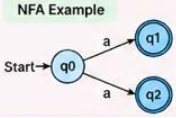
2 Finite Automata

- Deterministic finite automata (DFA)
- Nondeterministic finite automata (NFA)
- Equivalence of DFA and NFA
- Conversion between DFA and NFA
- Minimization of DFA
- Applications

DFA Example

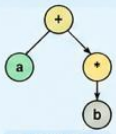


NFA Example



3 Regular Expressions

- Syntax and semantics of regular expressions
- Equivalence with finite automata
- Conversion: RE → NFA → DFA
- Nonregular languages
- Pumping lemma for regular languages



Syntax Tree of (a + b)*

Regular Expression Examples

(0 + 1)*	→ all binary strings
0*1*	→ any number of 0's followed by any number of 1's
(ab + ba)*	→ all strings of ab and ba

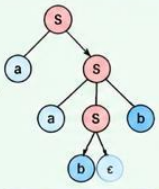
4 Context Free Languages

- Context free grammars (CFG)
- Derivations and parse trees
- Ambiguity in CFG
- Chomsky normal form (CNF)
- Pumping lemma for context free languages
- Applications of CFL

Example Grammar

S → aSb | ϵ

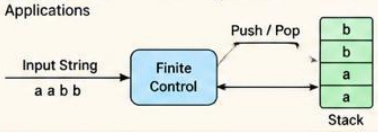
L = { $a^n b^n$ | n ≥ 0 }



Parse Tree for aabb

5 Pushdown Automata

- Pushdown automata (PDA)
- Components and formal definition
- Moves and instantaneous descriptions
- Acceptance by final state and empty stack
- Equivalence with context-free grammar
- Applications



6 Turing Machines

- Basic model of Turing machine
- Components: states, tape, transition function
- Variants of Turing machines
- Computing with Turing machines
- Languages recognized by Turing machines
- Relationship with other models



Transition Function

δ(q, 0) = (q1, 1, R)

(current state, read → next state, write, move)

7 Computability

- Church-Turing thesis
- Computable and non-computable functions
- Halting problem
- Diagonalization language
- Extensions and limitations

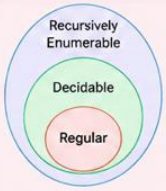
Halting Problem (Undecidable)

Given a program P and input w, determine whether P halts on w.

No Turing machine can solve the halting problem for all inputs.

8 Decidability

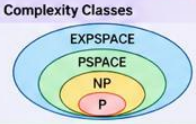
- Decidable languages
- Undecidable languages
- Halting problem
- Reduction techniques
- Recursively enumerable languages
- Properties of decidability



9 Complexity

- Time complexity
- Class P
- Class NP
- NP-completeness
- Cook's theorem
- Space complexity
- Savitch's theorem

Complexity Classes



Reference Books

1	Introduction to the Theory of Computation	Michael Sipser
2	Introduction to Theory of Computation	Anil Maheshwari, Michiel Smid